

torch.cholesky#

<https://pytorch.org/docs/stable/generated/torch.cholesky.html>

Description:

Computes the Cholesky decomposition of a symmetric positive-definite matrix `AAA` or for batches of symmetric positive-definite matrices. If `upper` is `True`, the returned matrix `U` is upper-triangular, and the decomposition has the form: If `upper` is `False`, the returned matrix `L` is lower-triangular, and the decomposition has the form: If `upper` is `True`, and `AAA` is a batch of symmetric positive-definite matrices, then the returned tensor will be composed of upper-triangular Cholesky factors of each of the individual matrices. Similarly, when `upper` is `False`, the returned tensor will be composed of lower-triangular Cholesky factors of each of the individual matrices.

Function Signature

`torch.cholesky(input, upper=False, *, out=None) → Tensor#`

Parameters

Keyword Arguments

Parameters

Keyword

out (Tensor, optional) – the output matrix

Code Examples

```
L = torch.linalg.cholesky(A)
```

```
U = torch.linalg.cholesky(A).mH
```

```
>>> a = torch.randn(3, 3)
>>> a = a @ a.mT + 1e-3 # make symmetric positive-definite
>>> l = torch.cholesky(a)
>>> a
tensor([[ 2.4112, -0.7486,  1.4551],
        [-0.7486,  1.3544,  0.1294],
        [ 1.4551,  0.1294,  1.6724]])
>>> l
tensor([[ 1.5528,  0.0000,  0.0000],
        [-0.4821,  1.0592,  0.0000],
        [ 0.9371,  0.5487,  0.7023]])
>>> l @ l.mT
tensor([[ 2.4112, -0.7486,  1.4551],
        [-0.7486,  1.3544,  0.1294],
        [ 1.4551,  0.1294,  1.6724]])
>>> a = torch.randn(3, 2, 2) # Example for batched input
>>> a = a @ a.mT + 1e-03 # make symmetric positive-definite
>>> l = torch.cholesky(a)
>>> z = l @ l.mT
>>> torch.dist(z, a)
tensor(2.3842e-07)
```

Notes & Warnings

WARNING

Warning `torch.cholesky()` is deprecated in favor of `torch.linalg.cholesky()` and will be removed in a future PyTorch release. `L = torch.cholesky(A)` should be replaced with `L = torch.linalg.cholesky(A)` `U = torch.cholesky(A, upper=True)` should be replaced with `U = torch.linalg.cholesky(A).mH` This transform will produce equivalent results for all valid (symmetric positive definite) inputs.

Detailed Documentation

Documentation

Computes the Cholesky decomposition of a symmetric positive-definite matrix `AAA` or for batches of symmetric positive-definite matrices.

If `upper` is `True`, the returned matrix `U` is upper-triangular, and the decomposition has the form:

If `upper` is `False`, the returned matrix `L` is lower-triangular, and the decomposition has the form:

If `upper` is `True`, and `AAA` is a batch of symmetric positive-definite matrices, then the returned tensor will be composed of upper-triangular Cholesky factors of each of the individual matrices. Similarly, when `upper` is `False`, the returned tensor will be composed of lower-triangular Cholesky factors of each of the individual matrices.

`torch.cholesky()` is deprecated in favor of `torch.linalg.cholesky()` and will be removed in a future PyTorch release.

`L = torch.cholesky(A)` should be replaced with

`U = torch.cholesky(A, upper=True)` should be replaced with

This transform will produce equivalent results for all valid (symmetric positive definite) inputs.

`input (Tensor)` – the input tensor `AAA` of size $(*,n,n)^{(*)}$, $(*,n,n)^{(*)}$ where $*$ is zero or more batch dimensions consisting of symmetric positive-definite matrices.

`upper (bool, optional)` – flag that indicates whether to return a upper or lower triangular matrix. Default: `False`

`out (Tensor, optional)` – the output matrix